Elementary & Intermediate Algebra

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A bag contains 8 red marbles, 6 green marbles, and 10 blue marbles. If 1 marble is chosen at random from the bag, what is the probability that the marble is NOT red?

To find the probability that the chosen marble is NOT red, determine the number of favorable outcomes and divide by the total number of outcomes. 1. Find the total number of marbles in the bag: $8 + 6 + 10 = 24$ 2. Find the number of marbles that are not red (the green and blue marbles): $6 + 10 = 16$ Alternatively, use the complement rule: $P(\text{not red}) = 1 - P(\text{red}) = 1 - \frac{8}{24} = 1 - \frac{1}{3} = \frac{2}{3}$ 3. Calculate the probability.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the standard (x, y) coordinate plane, the line segment with endpoints $A(1, -2)$ and $B(5, 6)$ is perpendicularly bisected by line ℓ. If the point $(9, k)$ lies on line ℓ, what is the value of k?

To find the value of k, determine the equation of the perpendicular bisector line ℓ in two main steps: 1. Find the midpoint of $\overline{AB}$: Line ℓ bisects segment $\overline{AB}$, so it passes through the midpoint M: $M = \left(\frac{1+5}{2}, \frac{-2+6}{2}\right) = (3, 2)$ 2. Find the slope of line ℓ: The slope of $\overline{AB}$ is: $m_{AB} = \frac{6-(-2)}{5-1} = \frac{8}{4} = 2$ Since line ℓ is perpendicular to $\overline{AB}$, its slope is the negative reciprocal of 2.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The function $f(x) = a\sin(bx) + c$ has a maximum value of 9, a minimum value of -1, and a period of $\frac{\pi}{2}$. If $a > 0$ and $b > 0$, what is the value of $a + b + c$?

To find the value of $a + b + c$, determine each parameter from the given properties of the sinusoidal function: 1. Amplitude a and Vertical Shift c: Since $-1 \le \sin(bx) \le 1$ and $a > 0$, the maximum and minimum values of $f(x)$ are: $\text{Maximum} = c + a = 9$ $\text{Minimum} = c - a = -1$ Adding these two equations gives: $2c = 8 \Rightarrow c = 4$ Substituting $c = 4$ into $c + a = 9$: $4 + a = 9 \Rightarrow a = 5$ 2. Frequency Parameter b: The period T of a sine function $\sin(bx)$ with.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what is the distance between the points $(-1, 2)$ and $(3, 5)$?

To find the distance between two points (x_1, y_1) and (x_2, y_2) in the standard (x, y) coordinate plane, apply the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ Substitute the coordinates of $(-1, 2)$ and $(3, 5)$: $x_2 - x_1 = 3 - (-1) = 4$ $y_2 - y_1 = 5 - 2 = 3$ Compute the distance: $d = \sqrt{4^2 + 3^2} = \sqrt{16 + 9} = \sqrt{25} = 5$ Common Misconceptions: - $\sqrt{7}$ comes from incorrectly subtracting rather than adding the squares: $\sqrt{4^2 - 3^2} = \sqrt{7}$. - 7 represents the.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, vector $\mathbf{u} = \langle 2, -5 \rangle$ and vector $\mathbf{v} = \langle k, 3 \rangle$, where $k > 0$. If the magnitude of the vector sum $\mathbf{u} + \mathbf{v}$ is $\sqrt{29}$, what is the value of k?

To find k, follow these steps: 1. Find the vector sum $\mathbf{u} + \mathbf{v}$: Add corresponding components: $\mathbf{u} + \mathbf{v} = \langle 2 + k, -5 + 3 \rangle = \langle k + 2, -2 \rangle$ 2. Express the magnitude of $\mathbf{u} + \mathbf{v}$: The magnitude of a vector $\langle x, y \rangle$ is given by $\sqrt{x^2 + y^2}$: $||\mathbf{u} + \mathbf{v}|| = \sqrt{(k + 2)^2 + (-2)^2} = \sqrt{(k + 2)^2 + 4}$ 3. Set up and solve the equation: Given that the magnitude is $\sqrt{29}$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Given the matrices $A = [2, -3; 1, 4]$ and $B = [-5, 1; 3, -2]$, what is the matrix $2A + B$?

To find $2A + B$, first perform scalar multiplication by multiplying each entry of matrix A by 2: $2A = 2[2, -3; 1, 4] = [2(2), 2(-3); 2(1), 2(4)] = [4, -6; 2, 8]$ Next, add matrix B to $2A$ by adding corresponding entries: $2A + B = [4, -6; 2, 8] + [-5, 1; 3, -2] = [4 + (-5), -6 + 1; 2 + 3, 8 + (-2)] = [-1, -5; 5, 6]$ Common errors: - Simply computing $A + B$ gives $[-3, -2; 4, 2]$. - Computing $2A - B$ gives.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the standard (x, y) coordinate plane, what are the coordinates of the midpoint of a line segment with endpoints $(-3, 8)$ and $(5, -2)$?

To find the midpoint (x_m, y_m) of a line segment with endpoints (x_1, y_1) and (x_2, y_2), use the midpoint formula: $x_m = \frac{x_1 + x_2}{2}, \quad y_m = \frac{y_1 + y_2}{2}$ Substitute the given coordinates $(-3, 8)$ and $(5, -2)$: $x_m = \frac{-3 + 5}{2} = \frac{2}{2} = 1$ $y_m = \frac{8 + (-2)}{2} = \frac{6}{2} = 3$ Thus, the midpoint is $(1, 3)$. Common errors: - Finding the difference instead of the average yields $\left(\frac{5 - (-3)}{2}, \frac{-2 - 8}{2}\right) = (4, -5)$. - Forgetting to divide t.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In the standard (x, y) coordinate plane, what is the midpoint of the line segment with endpoints $(3, -4)$ and $(-5, 8)$?

To find the midpoint of a line segment with endpoints (x_1, y_1) and (x_2, y_2), use the midpoint formula: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$ Substitute the coordinates of the endpoints $(3, -4)$ and $(-5, 8)$: $x_m = \frac{3 + (-5)}{2} = \frac{-2}{2} = -1$ $y_m = \frac{-4 + 8}{2} = \frac{4}{2} = 2$ Thus, the midpoint is $(-1, 2)$. Common errors: - $(-2, 4)$ results from adding coordinates without dividing by 2. - $(1, -2)$ results from sign errors during addition. - $(4, -6)$ results from.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A fair coin is tossed twice. Find $P(\text{two heads})$.

Because each coin flip is an independent event, apply the multiplication rule for independent probabilities: $P(\text{Head on 1st}) = \frac{1}{2}$ $P(\text{Head on 2nd}) = \frac{1}{2}$ $P(H_1 \cap H_2) = P(H_1) \times P(H_2) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4} = \mathbf{0.25}$ Sample Space Check: The sample space is $\{HH, HT, TH, TT\}$. Exactly 1 outcome out of 4 is HH, confirming $\frac{1}{4}$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A recipe is scaled from 2 to 5 servings. How many cups replace 4 cups?

Set up a direct proportion where x represents the number of cups for 5 servings: $\frac{2 \text{ servings}}{4 \text{ cups}} = \frac{5 \text{ servings}}{x \text{ cups}}$ Cross-multiply to solve for x: $2x = 5 \times 4 = 20 \Rightarrow x = \frac{20}{2} = \mathbf{10}$ cups Alternatively, calculate the scaling factor: $k = \frac{5}{2} = 2.5$. Then $4 \times 2.5 = \mathbf{10}$ cups.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A square has area 36. Find its perimeter.

Step 1: Find the side length s from area $A = s^2$: $s = \sqrt{36} = 6$ Step 2: Calculate the perimeter of the square $P = 4s$: $P = 4 \times 6 = \mathbf{24}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final result satisfies all domain restrictions, boundary conditions, and the original governing equation by direct substitution into the prompt.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has opposite 3 and hypotenuse 5. Find $\sin\theta$.

Recall the basic trigonometric ratio from SOH CAH TOA: $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{3}{5} = \mathbf{0.6}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line passes through $(1, 2)$ with slope 3. Find y when $x = 4$.

Method 1 (Point-Slope Form): $y - y_1 = m(x - x_1) \Rightarrow y - 2 = 3(x - 1)$ Substitute $x = 4$: $y - 2 = 3(4 - 1) = 3(3) = 9 \Rightarrow y = 9 + 2 = \mathbf{11}$ Method 2 (Rate of Change / Delta Method): $\Delta x = 4 - 1 = 3$. Since slope $m = \frac{\Delta y}{\Delta x} = 3$, $\Delta y = m \cdot \Delta x = 3 \cdot 3 = 9$. New $y = 2 + 9 = \mathbf{11}$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A random variable X has outcomes 7, 15, 19, 26 with $P(X = 7) = \frac{3}{12}, P(X = 15) = \frac{5}{12}, P(X = 19) = \frac{2}{12}, P(X = 26) = \frac{2}{12}$. Find $E(X)$.

$E(X) = 7\left(\frac{3}{12}\right) + 15\left(\frac{5}{12}\right) + 19\left(\frac{2}{12}\right) + 26\left(\frac{2}{12}\right) = \frac{31}{2}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 Functions & Graphs

For $f(x) = x^2$, find the average rate of change on $[1, 3]$.

The average rate of change on the closed interval $[a, b]$ is the slope of the secant line: $\text{Average Rate of Change} = \frac{f(b) - f(a)}{b - a}$ Compute function values at endpoints: $f(1) = 1^2 = 1, \quad f(3) = 3^2 = 9$ Substitute into the slope formula: $\frac{f(3) - f(1)}{3 - 1} = \frac{9 - 1}{2} = \frac{8}{2} = \mathbf{4}$

16 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Coordinate Geometry

If $\sin\theta = -\frac{9}{41}$ and θ is in Quadrant IV, find $\cos\theta$.

$\sin^2\theta + \cos^2\theta = 1 \Rightarrow \cos\theta = \pm\frac{40}{41}$. In Quadrant IV, cosine is positive: $\frac{40}{41}$. Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

17 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Plane Geometry

A USD 80 item is marked down 20%. Find the sale price.

Percent Decrease Multiplier Shortcut: A markdown of 20% means the customer pays $100\% - 20\% = 80\%$ of the original price. $\text{Sale Price} = 80 \times (1 - 0.20) = 80 \times 0.80 = \mathbf{USD64}$ Trap Warning: Do NOT subtract $USD20$ directly; 20% of $USD80$ is $USD16$, leaving $USD80 - USD16 = USD64$.

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18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Practice 2.2: What is the sum of the solutions of the quadratic equation $3x^2 - 12x - 5 = 0$?

By Vieta's formulas, for any quadratic equation $ax^2 + bx + c = 0$, the sum of the roots is $-\frac{b}{a}$. Here $a = 3$ and $b = -12$, so the sum of the solutions is: $-\frac{-12}{3} = \frac{12}{3} = 4$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

Practice 2.1: For what values of x is $|2x - 3| < 7$?

Rewrite as a compound inequality: $-7 < 2x - 3 < 7$. Add 3: $-4 < 2x < 10$. Divide by 2: $-2 < x < 5$. Solution set: $(-2, 5)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

What is the sum and product of the roots of $3x^2 - 15x + 7 = 0$?

By Vieta's formulas, sum $= -b/a = 15/3 =$ **5** and product $= c/a =$ **7/3**.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the linear equation for x: $\frac{2x-3}{4} - \frac{x+1}{3} = \frac{5}{12}$.

a. 7

b. 8 ✓

c. 9

d. 11

Multiply the entire equation by the common denominator 12:

$3(2x - 3) - 4(x + 1) = 5 \Rightarrow 6x - 9 - 4x - 4 = 5 \Rightarrow 2x - 13 = 5 \Rightarrow 2x = 18 \Rightarrow x = 9$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For what value of k does the equation $4(2x - 5) + k = 8x - 13$ have infinitely many solutions?

a. 7 ✓

b. 13

c. 20

d. -7

Expand left side: $8x - 20 + k = 8x - 13$. For identity, $-20 + k = -13 \Rightarrow k = 7$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the complete solution set for the compound inequality $-7 \leq 3x + 2 < 11$?

a. $-3 \leq x < 3$ ✓

b. $-5 \leq x < 4$

c. $-3 < x \leq 3$

d. $x \geq -3$

Subtract 2 from all parts: $-9 \leq 3x < 9$. Divide by 3: $-3 \leq x < 3$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A company sells widgets for 14 each with fixed costs of $1,800$ and variable costs of 5 per widget. How many widgets must be sold to make a profit of at least 900?

a. 200

b. 250

c. 300 ✓

d. 350

Profit equation: $14w - (5w + 1800) \geq 900 \Rightarrow 9w - 1800 \geq 900 \Rightarrow 9w \geq 2700 \Rightarrow w \geq 300$ widgets.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What are the solutions to the quadratic equation $2x^2 - 7x - 15 = 0$?

a. $x = 5$ **and** $x = -\frac{3}{2}$ ✓

b. $x = -5$ and $x = \frac{3}{2}$

c. $x = 3$ and $x = -\frac{5}{2}$

d. $x = 5$ and $x = 3$

Factor by grouping: $(2x + 3)(x - 5) = 0$. Thus $2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$, and $x - 5 = 0 \Rightarrow x = 5$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the sum of the solutions of the equation $3x^2 - 12x + 5 = 0$?

a. -4

b. 4 ✓

c. $\frac{5}{3}$

d. 12

By Vieta's formulas, the sum of solutions of $ax^2 + bx + c = 0$ is $-\frac{b}{a} = -\frac{-12}{3} = 4$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of c will the quadratic equation $x^2 - 10x + c = 0$ have exactly one distinct real solution?

a. 5

b. 10

c. 25 ✓

d. 100

Exactly one real solution requires discriminant $D = b^2 - 4ac = 0$. Here, $(-10)^2 - 4(1)(c) = 0 \Rightarrow 100 - 4c = 0 \Rightarrow c = 25$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Simplify the rational expression: $\frac{x^2 - 9}{2x^2 + 5x - 3}$ for all $x \neq \frac{1}{2}, -3$.

a. $\frac{x-3}{2x-1}$ ✓

b. $\frac{x+3}{2x+1}$

c. $\frac{x-3}{2x+1}$

d. $\frac{1}{2x-1}$

Factor numerator: $(x - 3)(x + 3)$. Factor denominator: $(2x - 1)(x + 3)$. Cancel common factor $(x + 3)$ to get $\frac{x-3}{2x-1}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the radical equation for all valid real values: $\sqrt{3x + 13} = x + 3$.

a. $x = 1$ **only** ✓

b. $x = -4$ only

c. $x = 1$ and $x = -4$

d. No real solutions

Square both sides: $3x + 13 = x^2 + 6x + 9 \Rightarrow x^2 + 3x - 4 = 0 \Rightarrow (x + 4)(x - 1) = 0$. Checking $x = 1$: $\sqrt{16} = 4$ (valid). Checking $x = -4$: $\sqrt{1} = -1$ (extraneous).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Use the quadratic formula to solve $x^2 + 13x - 6 = 0$. What is the sum of the roots?

a. -17

b. -10

c. -16

d. **-13** ✓

By Vieta's formulas, sum of roots $= -b/a = 0 - 13/1 = -13$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the real numbers x and y satisfy the equation $2x^2 + y^2 - 4x + 6y + 11 = 0$ what is the value of $x + y$?

a. -4

b. **-2** ✓

c. 2

d. 4

To solve for x and y from a single equation with real variables, we can complete the square for both x and y: 1. Group the x-terms and the y-terms: $(2x^2 - 4x) + (y^2 + 6y) + 11 = 0$ 2. Factor out the leading coefficient from the x-terms: $2(x^2 - 2x) + (y^2 + 6y) + 11 = 0$ 3. Complete the square inside each group: $2(x^2 - 2x + 1 - 1) + (y^2 + 6y + 9 - 9) + 11 = 0$ $2(x - 1)^2 - 2 + (y + 3)^2 - 9 + 11 = 0$ 4. Combine the constant terms: $2(x - 1)^2 + (y + 3)^2 = 0$ Because x and y are real numbe.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following expressions is equivalent to $(5x^2 - 2x + 7) - (3x^2 - 6x - 4)$?

a. $2x^2 - 8x + 3$

b. $2x^2 - 8x + 11$

c. $2x^2 + 4x + 3$

d. $2x^2 + 4x + 11$ ✓

To simplify the expression, distribute the negative sign across each term in the second set of parentheses: $-(3x^2 - 6x - 4) = -3x^2 + 6x + 4$ Now rewrite the entire expression and combine like terms: $(5x^2 - 2x + 7) - 3x^2 + 6x + 4$ Group terms with the same power of x: - Quadratic terms: $5x^2 - 3x^2 = 2x^2$ - Linear terms: $-2x + 6x = 4x$ - Constant terms: $7 + 4 = 11$ Combining these yields $2x^2 + 4x + 11$. Common misconceptions: - Failing to distribute the negative sign to $-6x$ yields $-2x - 6x = -8x$, p.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For all real values of x, which of the following expressions is equivalent to $(3x - 4)^2$?

a. $9x^2 - 24x + 16$ ✓

b. $9x^2 - 12x + 16$

c. $9x^2 - 16$

d. $9x^2 + 16$

To expand the squared binomial $(3x - 4)^2$, write it as the product of the binomial with itself: $(3x - 4)^2 = (3x - 4)(3x - 4)$ Apply the distributive property (FOIL method): - First: $(3x)(3x) = 9x^2$ - Outer: $(3x)(-4) = -12x$ - Inner: $(-4)(3x) = -12x$ - Last: $(-4)(-4) = +16$ Combine the like terms in the middle: $9x^2 - 12x - 12x + 16 = 9x^2 - 24x + 16$ Analysis of incorrect choices: - $9x^2 - 12x + 16$ accounts for only one of the cross-terms rather than doubling $2ab$. - $9x^2 - 16$ and $9x^2 + 16$ result.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve: $|x - 1| = 18$. What is the larger solution?

a. 13

b. **19** ✓

c. 24

d. 22

$x - 1 = 18$ or $x - 1 = -18$, giving $x = 19$ or $x = -17$. Larger: 19.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve: $x^2 - 4x + 3 = 0$. What is the larger root?

a. 1

b. 5

c. **3** ✓

d. 0

Factors as $(x - 1)(x - 3) = 0$, so $x = 1$ or $x = 3$. Larger root: 3.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve: $|x| = 10$. What is the positive solution?

a. 9

b. 12

c. **10** ✓

d. 13

$x = 10$ or $x = -10$; the positive solution is 10.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve: $|x - 10| = 7$. What is the sum of all solutions?

a. **20** ✓

b. 16

c. 24

d. 25

$x - 10 = \pm 7$, giving $x = 17$ or $x = 3$. Sum $= 20$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 Trigonometry

Evaluate $7^2 - 5^2$ using the difference of squares.

a. 19

b. 25

c. 29

d. **24** ✓

$7^2 - 5^2 = (7 - 5)(7 + 5) = (7 - 5) \times 12 = 24$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 7.01 Statistics & Probability

In the standard (x, y) coordinate plane below, the graph of the quadratic function $f(x) = 9 - x^2$ and the line $g(x) = x + 3$ intersect at points A and B, where A has a negative x-coordinate and B has a positive x-coordinate. Point C is the vertex of the parabola $y = f(x)$. What is the area, in square coordinate units, of ΔABC?

a. 12.5

b. **15** ✓

c. 22.5

d. 30

To find the area of ΔABC, we first determine the coordinates of its three vertices: A, B, and C. 1. **Vertex C**: The function $f(x) = 9 - x^2$ is a downward-opening parabola with its maximum at $x = 0$. $f(0) = 9 - 0^2 = 9$, so the vertex is $C(0, 9)$. 2. **Intersection Points A and B**: Set $f(x) = g(x)$: $9 - x^2 = x + 3$ Rearrange into standard quadratic form: $x^2 + x - 6 = 0$ Factor the quadratic equation: $(x + 3)(x - 2) = 0$ Thus, the x-coordinates of the intersection points are.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 8.01 Matrices, Vectors & Complex Numbers

In ΔABC shown below, the measure of $\angle A$ is 52° and the measure of $\angle B$ is 63°. What is the measure of $\angle C$?

a. 55°

b. 65° ✓

c. 75°

d. 115°

By the Triangle Angle Sum Theorem, the sum of the interior angle measures in any triangle is 180°: $m\angle A + m\angle B + m\angle C = 180^\circ$ Substitute the given angle measures into the equation: $52^\circ + 63^\circ + m\angle C = 180^\circ$ $115^\circ + m\angle C = 180^\circ$ $m\angle C = 180^\circ - 115^\circ = 65^\circ$ - 115° is an incorrect choice resulting from finding the sum of $\angle A$ and $\angle B$ without subtracting from.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Numbers, Percents & Ratios

Find the slope of the line through $(-6, 0)$ and $(-4, 4)$.

a. $\frac{1}{5}$

b. 2 ✓

c. $\frac{18}{25}$

d. $\frac{2}{5}$

Applying Coordinate Geometry principles, $m = \frac{4-0}{-4+6} = 2$. Therefore, the verified correct option is 2.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Functions & Graphs

A test consists of 25 multiple-choice questions. A student scored 80%. If the student correctly answered 4 more questions, what would the new percentage score be?

a. 84%

b. 88%

c. 92%

d. **96%** ✓

Initial correct: $0.80 \times 25 = 20$. New correct: $20 + 4 = 24$. New score: $24/25 = 96\%$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Coordinate Geometry

A circle has radius 9. What is the arc length of a 180° sector (in terms of π)?

a. 8π

b. 9π ✓

c. 11π

d. 18π

Arc length $= \frac{180}{360} \times 2\pi(9) = 9\pi$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Plane Geometry

A circle has diameter 24. What is its circumference (in terms of π)?

a. 24π ✓

b. 144π

c. 48π

d. 12π

Circumference $= 2\pi r = 2\pi(12) = 24\pi$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Trigonometry

A circle has radius 7. What is its circumference (in terms of π)?

a. 28π

b. 14π ✓

c. 7π

d. 49π

Circumference $= 2\pi r = 2\pi(7) = 14\pi$.

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